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A FAMILY OF DENSITIES DERIVED FROM THE THREE-PARAMETER DIRICHLET PROCESS

MATTHEW A. CARLTON,* *California Polytechnic State University*

Abstract

The traditional Dirichlet process is characterized by its distribution on a measurable partition of the state space—namely, the Dirichlet distribution. In this paper, we consider a generalization of the Dirichlet process and the family of multivariate distributions it induces, with particular attention to a special case where the multivariate density function is tractable.

Keywords: Dirichlet process; random measure; density function

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1. Introduction

We need to establish some basic notation. Hereafter we will use $\mathrm{Be}(\alpha, \beta)$ to denote the standard beta distribution with parameters α and β, and $D(\alpha_1, \ldots, \alpha_n)$ to denote the Dirichlet distribution on the n-simplex $\mathbb{S}_n$ with parameters $\alpha_1, \ldots, \alpha_n$.

We first consider a sequence W_n of independent and identically distributed (i.i.d.) $\mathrm{Be}(1, \theta)$ random variables and define a new sequence (P_n) as follows:

$$P_1 = W_1, \qquad P_n = W_n \prod_{i=1}^{n-1} (1 - W_i), \quad n > 1. \tag{1.1}$$

The construction in (1.1) is often called the *residual allocation model*, or RAM. Note that, by design, (P_n) is a probability sequence, i.e. the sequence satisfies the conditions (i) $P_n \geq 0$ for all n and (ii) $\sum_n P_n = 1$ almost surely. With this particular choice of W_n, the probability sequence (P_n) is said to have a $\mathrm{GEM}(\theta)$ distribution, named after Griffiths, Engen, and McCloskey.

Next, we use (P_n) to define a discrete random probability measure on a measure space $(\mathcal{X}, \mathcal{B})$. Let m be a measure on $(\mathcal{X}, \mathcal{B})$ with $\theta := m(\mathcal{X}) \in (0, \infty)$. Then we may define a probability measure μ on $(\mathcal{X}, \mathcal{B})$ by $\mu := \theta^{-1} m$. Let $(P_n) \sim \mathrm{GEM}(\theta)$ and, independent of (P_n), let $Y_1, Y_2, \ldots$ be i.i.d. μ, taking values in χ. We define a random measure $\mathfrak{P}$ on $(\mathcal{X}, \mathcal{B})$ by

$$\mathfrak{P} = \sum_{n=1}^{\infty} P_n \delta_{Y_n}. \tag{1.2}$$

That is, the random variables Y_n are the locations of the point masses of the measure $\mathfrak{P}$, and the probability sequence (P_n) gives their corresponding weights. This particular random measure is what is known as a *Dirichlet process* on $\mathcal{X}$, with 'parameter' m.

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* Postal address: Department of Statistics, California Polytechnic State University, San Luis Obispo, CA 93407, USA.
Email address: mcarlton@calpoly.edu

The Dirichlet process, first studied by Ferguson [3], is so named because of the following property: if $\mathfrak{P}$ is a Dirichlet process on $\mathcal{X}$ with parameter m and $\{B_1, \ldots, B_n\}$ is a measurable partition of $\mathcal{X}$, then

$$(\mathfrak{P}(B_1), \ldots, \mathfrak{P}(B_n)) \sim D(\alpha_1, \ldots, \alpha_n), \tag{1.3}$$

where $\alpha_i = m(B_i)$.

It should be noted that Ferguson uses (1.3) as the definition of a Dirichlet process in [3], and then proves the existence of such a measure using Kolmogorov's notion of cylinder sets. Ferguson also provides a construction in terms of a gamma process with independent increments. However, the constructive definition (1.2), which is arguably the simplest definition of a Dirichlet process, was first proposed by Sethuraman and Tiwari in [14]; the proof that this random measure $\mathfrak{P}$ satisfies (1.3) appears in [13].

Properties of the GEM(θ) distribution and of the Dirichlet process have been extensively researched over the past thirty-five years. For arguably the best summary of this history, see [10] and the bibliography therein. For our purposes, we need only mention one striking characteristic of the GEM(θ) distribution, which we describe now.

A *size-biased permutation* $(\tilde{P}_n)$ of a probability sequence (P_n) is defined as follows:

(i) $\mathrm{P}(\tilde{P}_1 = P_n \mid P_1, P_2, \ldots) = P_n$, and

(ii) for $j \geq 1$,

$$\mathrm{P}(\tilde{P}_{j+1} = P_n \mid \tilde{P}_1, \ldots, \tilde{P}_j, P_1, P_2, \ldots) = \frac{P_n}{1 - \tilde{P}_1 - \cdots - \tilde{P}_j} \mathbf{1}(P_n \neq \tilde{P}_1, \ldots, \tilde{P}_j).$$

McCloskey [5] proved that the GEM(θ) distribution is invariant under size-biased permutation, i.e. if $(P_n) \sim \mathrm{GEM}(\theta)$ and $(\tilde{P}_n)$ is a size-biased permutation of (P_n), then $(\tilde{P}_n) \stackrel{\mathrm{D}}{=} (P_n)$, where $\stackrel{\mathrm{D}}{=}$ denotes equality in distribution.

Pitman [8, 9], Perman [6] and Perman *et al.* [7] have found, in some sense, the 'maximal' family of probability sequences devisable under RAM which are invariant under size-biased permutation.

Theorem 1.1. (Pitman [9].) *Let W_n be a sequence of independent random variables such that $0 < W_n < 1$ almost surely, and suppose that (P_n) follows the RAM (1.1) with $\sum_n P_n = 1$ almost surely. Then (P_n) is invariant under size-biased permutation if and only if*

$$W_n \sim \mathrm{Be}(1 - \alpha, \theta + n\alpha) \tag{1.4}$$

for some pair (α, θ) with $\alpha \in [0, 1)$ and $\theta + \alpha > 0$.

For convenience, we will denote the space of acceptable parameters described in Theorem 1.1 by

$$\Theta := \{(\alpha, \theta) \in \mathbb{R}^2 \mid 0 \leq \alpha < 1,\ \theta > -\alpha\}.$$

We call the resulting sequence (P_n) in Theorem 1.1 the GEM(α, θ) distribution. This two-parameter family, in turn, generalizes the Dirichlet process (1.3). The random measure $\mathfrak{P}$ defined by (1.4), (1.1), and (1.2) is the *three-parameter Dirichlet process*, which we will denote by $\mathcal{D}(\mu; \alpha, \theta)$. When $\alpha = 0$, this becomes the original Dirichlet process with parameter $\theta\mu$.

Remark 1.1. We will treat μ and θ as 'separate' from this point onward rather than considering the amalgam $m = \theta\mu$ because the two pieces do not interact directly in the construction of the three-parameter Dirichlet process (θ helps to determine the weights P_n while μ determines the independent locations Y_n).

Our ultimate goal is, for the $\mathcal{D}(\mu;\alpha,\theta)$ process, to say as much as possible about the distribution of a random vector $(\mathfrak{P}(B_1),\ldots,\mathfrak{P}(B_n))$, analogous to (1.3), when $\alpha > 0$.

Before we begin, we introduce one more piece of notation. Let $\mathfrak{P} \in \mathcal{D}(\mu;\alpha,\theta)$ and let $\{B_1,\ldots,B_n\}$ be a measurable partition of $\mathcal{X}$ with $\mu(B_i) = p_i$ for $1 \le i \le n$. Let $\boldsymbol{R} = (\mathfrak{P}(B_1),\ldots,\mathfrak{P}(B_n))$. Then we will denote the distribution of $\boldsymbol{R}$ by $\mathrm{RS}(\boldsymbol{p};\alpha,\theta)$, where $\boldsymbol{p} = (p_1,\ldots,p_n)$. Likewise, for some $B \in \mathcal{B}$ with $\mu(B) = p$, we will denote the marginal distribution of $R = \mathfrak{P}(B)$ by $\mathrm{RS}(p;\alpha,\theta)$. (It is not immediately obvious that the distribution of $\boldsymbol{R}$ depends only on $\boldsymbol{p}$, α, and θ. However, this fact will become obvious later, and so we choose to use the above notation from here onward.)

The letters 'RS' are intended to abbreviate 'ratio-stable'. As shown in [6], the distribution in question can be formed by the ratio of stable random variables when $\theta = 0$, in the same way that the Dirichlet distribution (our $\alpha = 0$ case) can be formed by the ratio of gamma random variables.

Note that, when $\alpha = 0$, this RS distribution is the corresponding Dirichlet or beta distribution.

2. Some preliminary lemmas

To begin, we make a trivial observation about the Poisson–Dirichlet process, whose proof we will omit, which relies on the RAM description of the $\mathrm{GEM}(\alpha,\theta)$ distribution.

Lemma 2.1. *Suppose that $\mathfrak{P} \in \mathcal{D}(\mu;\alpha,\theta)$. Then*

$$\mathfrak{P} \stackrel{\mathrm{D}}{=} \delta_Y + (1-W)\mathfrak{P}^*,$$

where $W \sim \mathrm{Be}(1-\alpha,\theta+\alpha)$, $Y \sim \mu$, $\mathfrak{P}^* \in \mathcal{D}(\mu;\alpha,\theta+\alpha)$, and these are mutually independent.

Corollary 2.1. *Suppose that $\boldsymbol{R} \sim \mathrm{RS}(\boldsymbol{p};\alpha,\theta)$. Then*

$$\boldsymbol{R} \stackrel{\mathrm{D}}{=} W\boldsymbol{J} + (1-W)\boldsymbol{R}^*, \tag{2.1}$$

where $W \sim \mathrm{Be}(1-\alpha,\theta+\alpha)$, $\boldsymbol{R}^* \sim \mathrm{RS}(\boldsymbol{p};\alpha,\theta+\alpha)$, $\boldsymbol{J}$ is multinomial with parameters 1 and $\boldsymbol{p}$, and these are mutually independent.

Corollary 2.2. *Suppose that $R \sim \mathrm{RS}(p;\alpha,\theta)$. Then*

$$R \stackrel{\mathrm{D}}{=} WJ + (1-W)R^*,$$

where $W \sim \mathrm{Be}(1-\alpha,\theta+\alpha)$, $R^* \sim \mathrm{RS}(p;\alpha,\theta+\alpha)$, J is Bernoulli with parameter p, and these are mutually independent.

Our second lemma generalizes a result of Sethuraman [13].

Lemma 2.2. *Let $\mathcal{V}$ be a normed space over $\mathbb{R}$. For each $\lambda > 0$, let G_λ be a joint distribution on $\mathcal{V} \times \mathbb{R}$ with marginal distribution $\bar{G}_\lambda$ on $\mathbb{R}$. Suppose that, for some $\alpha \ge 0$,*

$$\prod_{n=1}^{\infty} W_n = 0 \quad \textit{almost surely}, \tag{2.2}$$

where $W_1, W_2, \ldots$ are independent and $W_n \sim \bar{G}_{\lambda+n\alpha}$. Then there exists at most one uniformly bounded family of distributions F_λ on $\mathcal{V}$ such that, for all $\lambda > 0$,

$$X \stackrel{\mathrm{D}}{=} V + WX^* \tag{2.3}$$

and X^* is independent of (V, W), where $X \sim F_\lambda$, $(V, W) \sim G_{\lambda+\alpha}$, and $X^* \sim F_{\lambda+\alpha}$.

Proof. Suppose that F_λ and F'_λ are uniformly bounded distributions on $\mathcal{V}$ which are not equal but which both satisfy (2.3). Let $X^{(\lambda)}$, $Y^{(\lambda)}$, and $(V^{(\lambda)}, W^{(\lambda)})$ be independent stochastic processes such that, for each fixed $\lambda > 0$, $X^{(\lambda)} \sim F_\lambda$, $Y^{(\lambda)} \sim F'_\lambda$, and $(V^{(\lambda)}, W^{(\lambda)}) \sim G_\lambda$. Let $(V_n^{(\lambda)}, W_n^{(\lambda)})$ be independent copies of $(V^{(\lambda)}, W^{(\lambda)})$ and define two sequences of stochastic processes $X_n^{(\lambda)}$ and $Y_n^{(\lambda)}$ by

$$X_1^{(\lambda)} = X^{(\lambda)}, \qquad X_{n+1}^{(\lambda)} = V_n^{(\lambda+\alpha)} + W_n^{(\lambda+\alpha)} X_n^{(\lambda+\alpha)},$$
$$Y_1^{(\lambda)} = Y^{(\lambda)}, \qquad Y_{n+1}^{(\lambda)} = V_n^{(\lambda+\alpha)} + W_n^{(\lambda+\alpha)} Y_n^{(\lambda+\alpha)}.$$

Observe that $X_n^{(\lambda)} \stackrel{\mathrm{D}}{=} X^{(\lambda)}$ and $Y_n^{(\lambda)} \stackrel{\mathrm{D}}{=} Y^{(\lambda)}$ for each fixed λ and for all n. Let $\|\cdot\|$ denote the norm of $\mathcal{V}$. Then

$$\begin{aligned}\|X_{n+1}^{(\lambda)} - Y_{n+1}^{(\lambda)}\| &= |W_n^{(\lambda+\alpha)}| \cdot \|X_n^{(\lambda+\alpha)} - Y_n^{(\lambda+\alpha)}\| = \cdots \\ &= \prod_{m=1}^{n} |W_m^{(\lambda+(n-m+1)\alpha)}| \cdot \|X_1^{(\lambda+n\alpha)} - Y_1^{(\lambda+n\alpha)}\| \\ &\le \prod_{m=1}^{n} |W_m^{(\lambda+m\alpha)}| c\end{aligned}$$

for some $c < \infty$ by the boundedness assumption. As $n \to \infty$, the right-hand side above converges to zero almost surely by assumption. This contradicts the supposition that the distributions F_λ and F'_λ are unequal and proves that, if the distribution F_λ is uniformly bounded and satisfies (2.3), then it is unique.

Next, we present a direct proof that our particular choice of W_n satisfies the condition (2.2).

Lemma 2.3. *For $(\alpha, \theta) \in \Theta$, let W_n, $n \ge 1$, be independent* $\mathrm{Be}(1-\alpha, \theta + n\alpha)$ *random variables. Then*

$$\prod_{n=1}^{\infty} (1 - W_n) = 0 \quad \textit{almost surely}$$

Proof. If $\alpha = 0$, then the above result is trivially true since the W_n are then i.i.d. and concentrated on $(0, 1)$. We proceed now under the assumption that $0 < \alpha < 1$.

Observe that $\prod_{n=1}^{\infty}(1 - W_n) = 0$ is equivalent to $\sum_{n=1}^{\infty} -\log(1 - W_n) = +\infty$. By the Kolmogorov 0–1 law, this series either diverges almost surely or converges almost surely. To show that this series diverges almost surely, as desired, it suffices by the Kolmogorov three series theorem to show that the series $\sum_{n=1}^{\infty} \mathrm{E}\, Y_n$ diverges, where

$$Y_n = -\log(1 - W_n)\mathbf{1}(|-\log(1 - W_n)| \le A)$$

for some $A > 0$. Let $a = 1 - \mathrm{e}^{-A} \in (0, 1)$. Then

$$\begin{aligned}\mathrm{E}\, Y_n &= \mathrm{E}[-\log(1 - W_n)\mathbf{1}(|-\log(1 - W_n)| \le A)] \\ &= \mathrm{E}[-\log(1 - W_n)\mathbf{1}(W_n \le a)] \\ &\ge \mathrm{E}[W_n \mathbf{1}(W_n \le a)] \\ &= \mathrm{E}\, W_n - \mathrm{E}[W_n \mathbf{1}(W_n > a)].\end{aligned}$$

Using the fact that $W_n/a > 1$ on $\{W_n > a\}$, the second term above can be bounded as follows:

$$\mathrm{E}[W_n \mathbf{1}(W_n > a)] \le \mathrm{E}\left[\frac{W_n^2}{a}\mathbf{1}(W_n > a)\right] \le \frac{\mathrm{E}\, W_n^2}{a} \le \frac{(1-\alpha)(2-\alpha)}{a(1-\alpha+\theta+n\alpha)^2}.$$

Combining, we have that

$$\mathrm{E}\, Y_n \geq \frac{1-\alpha}{1-\alpha+\theta+n\alpha} - \frac{(1-\alpha)(2-\alpha)}{a(1-\alpha+\theta+n\alpha)^2}.$$

This last difference of terms is clearly not summable, since the first term is $\mathcal{O}(1/n)$ while the second term is $\mathcal{O}(1/n^2)$. (Note that this also implies that the difference is nonnegative for sufficiently large n, so that the comparison test is applicable here.) Hence, by comparison, $\sum \mathrm{E}\, Y_n$ also diverges.

These three lemmas will allow us to take the approach used in the next section: suppose that we find a 'candidate' density for the random vector $\boldsymbol{R} \sim \mathrm{RS}(\boldsymbol{p}; \alpha, \theta)$. Most of the hypotheses of Lemma 2.2 are automatically satisfied by this vector, thanks to Lemma 2.1; the hypothesis in (2.2) is verified by Lemma 2.3. Hence, if we can show that our candidate density satisfies (2.1), then it must necessarily be correct. The same holds, of course, for the univariate case $R \sim \mathrm{RS}(p; \alpha, \theta)$.

3. A density formula

We are now ready to attempt to find the aforementioned density function. Unfortunately, a tractable formula has not been found for general α; it can be shown that, for $\alpha > 0$, the GEM(α, θ) distribution relates to certain stable distributions with index α (see [6] for this result). As is well known, tractable formulae exist only when $\alpha = \frac{1}{2}$, 1, or 2; only the first of these falls in our range of acceptable values. Hence, we present here the density of the RS $(\boldsymbol{p}; \alpha, \theta)$ distribution only for the special case $\alpha = \frac{1}{2}$.

Theorem 3.1. *Let $\boldsymbol{R} \sim \mathrm{RS}(\boldsymbol{p}; \frac{1}{2}, \theta)$ for some $\boldsymbol{p} \in \mathbb{S}_n$ and $\theta > -\frac{1}{2}$. Then the density of $\boldsymbol{R}$ is*

$$f_{\boldsymbol{R}}(\boldsymbol{r}) = \frac{p_1 \cdots p_n \Gamma(\theta + n/2)}{\pi^{(n-1)/2}\Gamma(\theta + 1/2)} \frac{r_1^{-3/2} \cdots r_n^{-3/2}}{(p_1^2/r_1 + \cdots + p_n^2/r_n)^{\theta+n/2}} \mathbf{1}(\boldsymbol{r} \in \mathbb{S}_n), \tag{3.1}$$

where $\boldsymbol{r} = (r_1, \ldots, r_n)$.

Detailed descriptions of the compelling reasons for this conjecture can be found in [1].

Proof. By Section 2, we need only show that the density (3.1) satisfies (2.1). To avoid notational confusion, let $\boldsymbol{X}$ denote $\boldsymbol{R}^*$ in (2.1). Let $\boldsymbol{e}_j$ denote the jth standard unit vector in $\mathbb{R}^n$. Conditional on $\boldsymbol{J} = \boldsymbol{e}_j$,

$$\boldsymbol{R} \mid \{\boldsymbol{J} = \boldsymbol{e}_j\} \stackrel{\mathrm{D}}{=} W\boldsymbol{e}_j + (1 - W)\boldsymbol{X}.$$

Let $\boldsymbol{Y}$ denote this conditional version of $\boldsymbol{R}$. Because $\boldsymbol{X}$ and W are independent,

$$f_{\boldsymbol{X},W}(\boldsymbol{x}, w) = f_W(w) f_{\boldsymbol{X}}(\boldsymbol{x}) = \frac{w^{-1/2}(1-w)^{\theta-1/2}}{B(1/2, \theta + 1/2)} f_{\boldsymbol{X}}(\boldsymbol{x}),$$

where $B(a, b)$ denotes the beta function $\Gamma(a)\Gamma(b)/\Gamma(a + b)$. Then, since the substitution $y_i = w\delta_{ij} + (1 - w)x_i$ prescribed above has Jacobian

$$\frac{\partial(x_1, \ldots, x_{n-1})}{\partial(y_1, \ldots, y_{n-1})} = (1 - w)^{1-n},$$

we have

$$f_{\boldsymbol{Y},W}(\boldsymbol{y}, w) = \frac{w^{-1/2}(1-w)^{\theta-1/2}}{B(1/2, \theta+1/2)} f_{\boldsymbol{X}}\left(\frac{y_1}{1-w}, \ldots, \frac{y_j - w}{1-w}, \ldots, \frac{y_n}{1-w}\right)(1-w)^{1-n}.$$

The singular density $f_{\boldsymbol{X}}$ lies on the set $\{x_i \geq 0,\ x_1 + \cdots + x_n = 1\}$, which under the above substitution moves to $\{y_i \geq \delta_{ij} w,\ y_1 + \cdots + y_n = 1\}$. Thus, integrating out w gives

$$f_{\boldsymbol{Y}}(\boldsymbol{y}) = \int_0^{y_j} \frac{w^{-1/2}(1-w)^{\theta+1/2-n}}{B(1/2, \theta+1/2)} f_{\boldsymbol{X}}\left(\frac{y_1}{1-w}, \ldots, \frac{y_j - w}{1-w}, \ldots, \frac{y_n}{1-w}\right) \mathrm{d}w.$$

We form the unconditional distribution of $\boldsymbol{R}$ by summing across all possible values of $\boldsymbol{J}$ and obtain

$$f_{\boldsymbol{R}}(\boldsymbol{r}) = \sum_{j=1}^{n} p_j \int_0^{r_j} \frac{w^{-1/2}(1-w)^{\theta+1/2-n}}{B(1/2, \theta+1/2)} f_{\boldsymbol{X}}\left(\frac{\boldsymbol{r} - w\boldsymbol{e}_j}{1-w}\right) \mathrm{d}w. \tag{3.2}$$

Substitute our proposed density for $\boldsymbol{X}$, i.e. shift θ to $\theta + \frac{1}{2}$ in (3.1). The substitution $w = r_j t$ and some straightforward algebra give

$$\frac{p_1 \cdots p_n \Gamma(\theta + (n+1)/2)}{\pi^{(n-1)/2}\Gamma(\theta+1)B(1/2, \theta+1/2)} \prod_{i \neq j} r_i^{-3/2} \int_0^1 \frac{r_j^{-1} t^{-1/2}(1-t)^{\theta+n/2-1}}{(p_j^2/r_j + (\sum_{i \neq j} p_i^2/r_i)(1-t))^{\theta+(n+1)/2}} \mathrm{d}t$$

for the jth integral in the sum (3.2). If we recognize this integral as a relative of the Gauss hypergeometric function (see (4.1) below), our expression quickly simplifies to

$$\frac{p_1 \cdots p_n \Gamma(\theta + (n+1)/2) B(\theta + n/2, 1/2)}{\pi^{(n-1)/2}\Gamma(\theta+1)B(1/2, \theta+1/2)} \prod_{i=1}^{n} r_i^{-3/2} \frac{r_j}{p_j(p_1^2/r_1 + \cdots + p_n^2/r_n)^{\theta+n/2}}.$$

Summing these against the weights p_j yields

$$\begin{aligned} f_{\boldsymbol{R}}(\boldsymbol{r}) &= \frac{p_1 \ldots p_n \Gamma(\theta + (n+1)/2) B(\theta + n/2, 1/2)}{\pi^{(n-1)/2}\Gamma(\theta+1)B(1/2, \theta+1/2)} \frac{r_1^{-3/2} \cdots r_n^{-3/2}}{(p_1^2/r_1 + \cdots + p_n^2/r_n)^{\theta+n/2}} \\ &= \frac{p_1 \ldots p_n \Gamma(\theta + n/2)}{\pi^{(n-1)/2}\Gamma(\theta+1/2)} \frac{r_1^{-3/2} \ldots r_n^{-3/2}}{(p_1^2/r_1 + \cdots + p_n^2/r_n)^{\theta+n/2}}, \end{aligned}$$

exactly as claimed.

Technically, we have not shown that (3.1) is a density function, i.e. that we have the correct constant of integration. We handle this now.

Lemma 3.1. *We have*

$$\int_{\mathbb{S}_n} p_1 \cdots p_n \frac{x_1^{-3/2} \cdots x_n^{-3/2}}{(p_1^2/x_1 + \cdots + p_n^2/x_n)^{\theta+n/2}} \mathrm{d}\boldsymbol{x} = \frac{\pi^{(n-1)/2}}{\Gamma(\theta+1/2)} \Gamma(\theta + n/2). \tag{3.3}$$

Proof. Note that

$$\begin{aligned} \int_0^\infty \sigma^{-a-1} \mathrm{e}^{-b/\sigma} \mathrm{d}\sigma &= \int_0^\infty u^{a+1} \mathrm{e}^{-bu} u^{-2} \mathrm{d}u \\ &= \int_0^\infty u^{a-1} \mathrm{e}^{-bu} \mathrm{d}u \\ &= \Gamma(a) b^{-a}. \end{aligned}$$

Let $a = \theta + n/2$ and $b = \sum_1^n p_i^2/x_i$. Then the integral in (3.3) equals

$$\int_{\mathbb{S}_n} p_1 \cdots p_n \frac{x_1^{-3/2} \cdots x_n^{-3/2}}{b^a}\,\mathrm{d}\boldsymbol{x} = \frac{1}{\Gamma(a)2^a}\int_{\mathbb{S}_n} p_1 \cdots p_n x_1^{-3/2} \cdots x_n^{-3/2}\Gamma(a)\left(\frac{b}{2}\right)^{-a}\mathrm{d}\boldsymbol{x}$$
$$= \frac{p_1 \cdots p_n}{\Gamma(a)2^a}\int_{\mathbb{S}_n}\int_0^\infty x_1^{-3/2} \cdots x_n^{-3/2}\sigma^{-a-1}\mathrm{e}^{-b/2\sigma}\,\mathrm{d}\sigma\,\mathrm{d}\boldsymbol{x}.$$

Now let $s_j = \sigma x_j$. Keep in mind that x_n is merely a symbol; the exterior integral ranges over a nonsingular space in $\mathbb{R}^{n-1}$. Note that $s_1 + \cdots + s_n = \sigma$, a convenient notation which we will preserve despite transforming from $(x_1, \ldots, x_{n-1}, \sigma)$ to $(s_1, \ldots, s_n)$. The Jacobian of the said transformation is the reciprocal of its inverse, which we can easily compute to be σ^{n-1}. Hence, the Jacobian here is σ^{1-n}, and the integral becomes

$$\frac{p_1 \cdots p_n}{\Gamma(a)2^a}\int_{\mathbb{R}^n} (\sigma^{-1}s_1)^{-3/2} \cdots (\sigma^{-1}s_n)^{-3/2}\sigma^{-a-1}\mathrm{e}^{-b/2\sigma}\sigma^{1-n}\,\mathrm{d}\boldsymbol{s}$$
$$= \frac{p_1 \cdots p_n}{\Gamma(a)2^a}\int_{\mathbb{R}^n} \sigma^{-\theta}\prod_{j=1}^n s_j^{-3/2}\mathrm{e}^{-p_j^2/2s_j}\,\mathrm{d}\boldsymbol{s}$$
$$= \frac{\pi^{n/2}}{\Gamma(\theta + n/2)2^\theta}\int_{\mathbb{R}^n} \sigma^{-\theta}\prod_{j=1}^n \frac{p_j}{\sqrt{2\pi}}s_j^{-3/2}\mathrm{e}^{-p_j^2/2s_j}\,\mathrm{d}\boldsymbol{s}.$$

We recognize the product within the integral as the joint density of n independent, positive, strictly-stable random variables with index $\frac{1}{2}$ and scale parameters p_j. Hence, the previous expression equals

$$\frac{\pi^{n/2}}{\Gamma(\theta + n/2)2^\theta}\,\mathrm{E}[\sigma^{-\theta}],$$

where σ is the sum of these stable random variables. By additivity of the stable distribution, σ is likewise stable, with index $\frac{1}{2}$ and scale parameter $p_1 + \cdots + p_n = 1$, so the probability density function of σ is

$$f_\sigma(\sigma) = \frac{1}{\sqrt{2\pi}}\sigma^{-3/2}\mathrm{e}^{-\sigma/2}.$$

Finally, we get

$$\frac{\pi^{n/2}}{\Gamma(\theta + n/2)2^\theta}\,\mathrm{E}[\sigma^{-\theta}] = \frac{\pi^{n/2}}{\Gamma(\theta + n/2)2^\theta}\int_0^\infty \frac{1}{\sqrt{2\pi}}\sigma^{-3/2-\theta}\mathrm{e}^{-\sigma/2}\,\mathrm{d}\sigma$$
$$= \frac{\pi^{(n-1)/2}}{\Gamma(\theta + n/2)2^{\theta+1/2}}\int_0^\infty \sigma^{-(\theta+1/2)-1}\mathrm{e}^{-\sigma/2}\,\mathrm{d}\sigma$$
$$= \frac{\pi^{(n-1)/2}}{\Gamma(\theta + n/2)2^{\theta+1/2}}\Gamma(\theta + \tfrac{1}{2})2^{\theta+1/2}$$
$$= \frac{\pi^{(n-1)/2}\Gamma(\theta + 1/2)}{\Gamma(\theta + n/2)}.$$

Remark 3.1. The univariate analogue for (3.1) is

$$\frac{p(1-p)}{B(1/2, \theta + 1/2)}\,\frac{r^{\theta-1/2}(1-r)^{\theta-1/2}}{(p^2(1-r) + (1-p)^2 r)^{\theta+1}}. \tag{3.4}$$

4. Moment formulae and other miscellany

A natural question which arises for any new density is whether we can formulate the moments of that density in a simple way. Naturally, we can make an attempt at integration against either (3.1) or (3.4), but these prove extremely difficult. We offer an alternative method here, which happens to work for general $(\alpha, \theta) \in \Theta$.

Referring to (1.2), we can write a random variable $R \sim \mathrm{RS}(p, \alpha, \theta)$ as

$$R = \sum_{n=1}^{\infty} P_n J_n,$$

where $(P_n) \sim \mathrm{GEM}(\alpha, \theta)$ and, independent of (P_n), the J_n are i.i.d. Bernoulli random variables with parameter p. We can attack this sum and its powers directly, thanks in part to the following simple lemma. The proof can be found in [12].

Lemma 4.1. (Ewens [2], Pitman and Yor [12].) *Suppose that* $(P_n) \sim \mathrm{GEM}(\alpha, \theta)$. *Then, for any measurable function* $f(t)$ *which is* $O(t)$ *as* $t \to 0$,

$$\mathrm{E}\left[\sum_{n=1}^{\infty} f(P_n)\right] = \mathrm{E}\left[\frac{f(W)}{W}\right],$$

where $W \sim \mathrm{Be}(1-\alpha, \theta+\alpha)$.

Brute-force computations yield the following results.

Lemma 4.2. *Suppose that* $\mathfrak{P} \in \mathcal{D}(\mu; \alpha, \theta)$ *on a measure space* $(\mathcal{X}, \mathcal{B})$, *and that* $B_1, B_2 \in \mathcal{B}$, $B_1 \cap B_2 = \varnothing$, *and* $\mu(B_j) = p_j$. *Let* $R_j = \mathfrak{P}(B_j)$, $j = 1, 2$. *Then, for all* $(\alpha, \theta) \in \Theta$,

$$\begin{aligned}
\mathrm{E}[R_1 - p_1] &= 0, \\
\mathrm{E}[(R_1 - p_1)^2] &= p_1(1-p_1)\frac{1-\alpha}{\theta+1}, \\
\mathrm{E}[(R_1 - p_1)^3] &= p_1(1-p_1)(1-2p_1)\frac{(1-\alpha)(2-\alpha)}{(\theta+1)(\theta+2)}, \\
\mathrm{E}[(R_1 - p_1)(R_2 - p_2)] &= -p_1 p_2 \frac{1-\alpha}{\theta+1}, \\
\mathrm{E}[(R_1 - p_1)^2(R_2 - p_2)] &= p_1 p_2 (2p_1 - 1)\frac{(1-\alpha)(2-\alpha)}{(\theta+1)(\theta+2)}.
\end{aligned}$$

Proof. For notational ease, we will omit the subscript on R in these computations. We have

$$\begin{aligned}
\mathrm{E}\, R &= \mathrm{E}\left[\sum_{n=1}^{\infty} P_n J_n\right] = \sum_{n=1}^{\infty} \mathrm{E}[P_n J_n] = \sum_{n=1}^{\infty} \mathrm{E}\, P_n \,\mathrm{E}\, J_n \\
&= \sum_{n=1}^{\infty} \mathrm{E}\, P_n \cdot p = p \sum_{n=1}^{\infty} \mathrm{E}\, P_n = p\,\mathrm{E}\left[\sum_{n=1}^{\infty} P_n\right] = p \cdot 1 = p
\end{aligned}$$

and

$$\begin{aligned}
\mathrm{E}[R^2] &= \mathrm{E}\left[\left(\sum_{n=1}^{\infty} P_n J_n\right)^2\right] = \mathrm{E}\left[\sum_{n=1}^{\infty}\sum_{m=1}^{\infty} P_m P_n J_m J_n\right] = \sum_{n=1}^{\infty}\sum_{m=1}^{\infty} \mathrm{E}[P_m P_n]\,\mathrm{E}[J_m J_n] \\
&= \sum_{m\neq n} \mathrm{E}[P_m P_n]\,\mathrm{E}\,J_m\,\mathrm{E}\,J_n + \sum_{n=1}^{\infty} \mathrm{E}\,P_n^2\,\mathrm{E}\,J_n^2 \\
&= p^2 \sum_{m\neq n} \mathrm{E}[P_m P_n] + p\sum_{n=1}^{\infty} \mathrm{E}\,P_n^2 \\
&= p^2 \sum_{n=1}^{\infty}\sum_{m=1}^{\infty} E[P_m P_n] + (p-p^2)\sum_{n=1}^{\infty} \mathrm{E}\,P_n^2 \\
&= p^2\left(\mathrm{E}\left[\sum_{n=1}^{\infty} P_n\right]\right)^2 + p(1-p)\,\mathrm{E}\left[\frac{W^2}{W}\right] \quad \text{(by Lemma 4.1)} \\
&= p^2 + p(1-p)\frac{1-\alpha}{\theta+1}.
\end{aligned}$$

The first two results immediately follow, and the others are similar (see [1] for details).

It is curious to note that these equations show that

$$\rho(R_1, R_2) = -\sqrt{\frac{p_1}{1-p_1}\frac{p_2}{1-p_2}},$$

i.e. the correlation between any two elements of the random vector $\boldsymbol{R}$ does not depend on θ or α.

Unfortunately, higher moments cannot be computed by this method; terms not covered by Lemma 4.1 appear when the exponent is four or more. We can, however, offer a general method for computing all moments of the density (3.4). First, note that, when $p = \frac{1}{2}$, (3.4) actually simplifies to the $\mathrm{Be}(\theta + \frac{1}{2}, \theta + \frac{1}{2})$ distribution. Otherwise, we can recognize the relevant integrand,

$$\frac{p(1-p)}{B(1/2, \theta+1/2)}\frac{r^{\theta+n-1/2}(1-r)^{\theta-1/2}}{[p^2(1-r)+(1-p)^2 r]^{\theta+1}}$$

as a relative of the Gauss hypergeometric function, which is given by the integral equation

$$H(a,b,c,z) = \frac{1}{B(b,c-b)}\int_0^1 t^{b-1}(1-t)^{c-b-1}(1-tz)^{-a}\,\mathrm{d}t \tag{4.1}$$

for $\mathrm{Re}(c) > \mathrm{Re}(b) > 0$.

We catalogue here some of the known properties of the Gauss hypergeometric function which are used in the proof of Lemma 4.4; see, for example, [15].

Lemma 4.3. *Let $H(a,b,c,z)$ be as in (4.1). Then*

(i)

$$H(a,b,c,z) = (1-z)^{-a} H\left(a, c-b, c, \frac{z}{z-1}\right);$$

(ii)
$$c(c-1)(z-1)H(a,b,c-1,z)+c[c-1-(2c-a-b-1)z]H(a,b,c,z) \\ +(c-a)(c-b)zH(a,b,c+1,z)=0;$$

(iii) *if* $c=a+b-\frac{1}{2}$, *then*
$$H(a,b,c,z)=(1-z)^{-1/2}H\left(2a-1,2b-1,c,\frac{1-\sqrt{1-z}}{2}\right);$$

(iv)
$$H(a,b,c,z)=(1-z)^{c-a-b}H(c-a,c-b,c,z);$$

(v) *for* $m\in\mathbb{N}$,
$$H(-m,b,c,z)=\sum_{k=0}^{m}(-1)^k\binom{m}{k}\frac{(b)_k}{(c)_k}z^k,$$
where $(x)_n=x(x+1)\cdots(x+n-1)=\Gamma(x+n)/\Gamma(x)$.

Using Lemma 4.3 and some algebra, we arrive at the following result.

Lemma 4.4. *Suppose that* $R\sim \mathrm{RS}(p;\frac{1}{2},\theta)$ *for some* $\theta>-\frac{1}{2}$ *and* $p\neq\frac{1}{2}$, *and let* $m_n=\mathrm{E}\,R^n$. *Then*

$$m_0=1 \qquad (p\neq 0),$$
$$m_1=p,$$

and, for $n\geq 2$,
$$m_n=\frac{(3-2n-2\theta)p^2m_{n-2}+(3-6p+(4\theta-2)p^2+2n(p^2+2p-1))m_{n-1}}{2(\theta+n-1)(2p-1)}.$$

5. Bayesian analysis and the posterior distribution

Ferguson [3] initially proposed the Dirichlet process for use in nonparametric Bayesian analysis. In particular, he established the following result for the original case (i.e. $\alpha=0$ and $m:=\theta\mu$): if $\mathfrak{P}$ is a Dirichlet process with measure parameter m and $X_1,\ldots,X_n$ are a random sample from $\mathfrak{P}$, then the conditional distribution of $\mathfrak{P}$ given $X_1,\ldots,X_n$ is a Dirichlet process with measure parameter $m+\sum\delta_{X_i}$. That is, if the prior distribution of $\mathfrak{P}$ is a Dirichlet process, then so is its posterior distribution given the observations $X_1,\ldots,X_n$, with the measure parameter updated appropriately.

Unfortunately, this result does not extend to the $\alpha>0$ case. The following theorem answers the question of the posterior distribution, provided that the 'prior' probability measure μ is nonatomic. This theorem first appeared, without proof, in [10]; the proof appears in [1].

Theorem 5.1. *Suppose that* $\mathfrak{P}\in\mathcal{D}(\mu;\alpha,\theta)$ *with* μ *nonatomic. Let* $X_1,\ldots,X_n$ *be a sample from* $\mathfrak{P}$. *Let* K *be the number of distinct* X_i, X'_j *be the* j*th distinct* X_i, *and* n_j *be the number of* X_i *equal to* X'_j. *Then*
$$\mathfrak{P}\mid X_1,\ldots,X_n \overset{\mathrm{D}}{=} \sum_{j=1}^{K}W_j\delta_{X'_j}+W_{K+1}\mathfrak{P}_K, \tag{5.1}$$
where $\mathfrak{P}_K\in\mathcal{D}(\mu;\alpha,\theta+K\alpha)$, $(W_1,\ldots,W_{K+1})\sim D(n_1-\alpha,\ldots,n_K-\alpha,\theta+K\alpha)$, *and these are independent.*

In Theorem 5.1, we need to assume that the measure μ is nonatomic (i.e. that $\mu(\{x\}) = 0$ for all $x \in \mathcal{X}$) so that μ-random variables are distinct almost surely. We briefly consider here the case when μ is atomic (i.e. μ has atoms).

First, the density formulae of Sections 3 hold regardless of whether μ has atoms. In fact, we can extract information about $\mathfrak{P} \in \mathcal{D}(\mu; \alpha, \theta)$ applied to an atom of μ for these formulae. For example, suppose that $a \in \mathcal{X}$ and $\mu(\{a\}) = p > 0$. Then (3.4) gives the density of $\mathfrak{P}(\{a\})$ when $\alpha = \frac{1}{2}$, namely RS $(p; \frac{1}{2}, \theta)$.

Second, note that Theorem 5.1 still holds for atomic measures if the sample $X_1, \ldots, X_n$ avoids these atoms. However, if μ is purely atomic (i.e. $\mu = \sum_{a \in \mathcal{A}} p_a \delta_a$ for some $\mathcal{A} \subset \mathcal{X}$), then Theorem 5.1 offers us no help.

The current work of Pitman may be of some assistance. Hansen and Pitman [4] show that mixtures of the form (5.1) still exist for atomic measures but involve the random sets

$$\{\{i \leq n \mid X_i = a\} \mid a \text{ is an atom of } \mu\}.$$

Corollary 33 of [11] gives the Mellin transform of the density of $\mathfrak{P}(\{a\})$ for the case $\theta = \alpha \in (0, 1)$. Unfortunately, few other results exist.

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